

Region Growing & Region Splitting

segmentation :

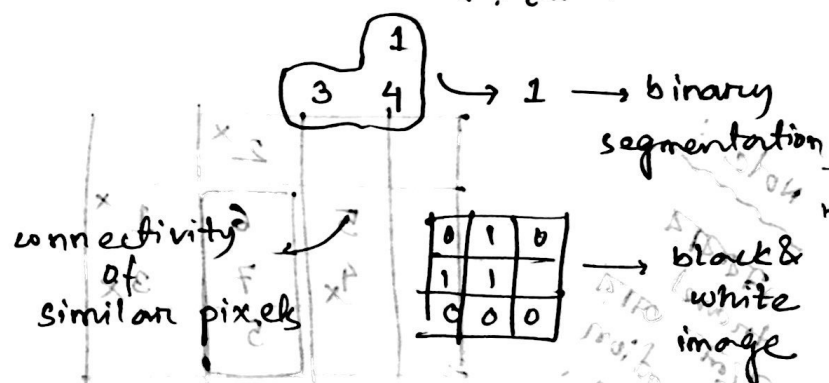
Binary segmentation
color

0	1	2
3	4	5
6	7	8

→ 9 different regions

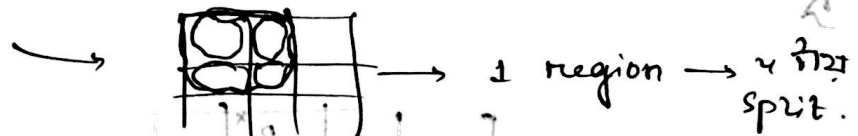
→ one criteria or basis a segment a jati

Given : threshold / cut off = value → region compare



→ Region growing / merging

→ splitting



Region growing

Q1 →

	0	1	2	3
0	0	1	2	0
1	2	5	6	1
2	1	4	7	3
3	0	2	5	1

(2, 2) → initial point.

$T = 2$ → Threshold value

pixel value aur diff.
2 aur tak hai na,

(diff ≤ 2)

↓
color diff

4. connectivity (+)

7 - 6 = 1 ≤ 2

4 ← 7 → 3
↓
5

✓ 7 - 5 = 2 ≤ 2

✗ 7 - 4 = 3 > 2

✗ 7 - 3 = 4 > 2

		2 ^x	
	5	6	1 ^x
	4 ^x	7	3 ^x
		5	

Now work with 6 → 7 वाप → as आया
consider करेगा !

	1	2 ^x	
2	5	6	1 ^x
1	4 ^x	7	3 ^x
		5	

Now work with 5 → diff ≥ 2 → No more
growing.

(4 आया
consider done → so
निरा ना)

Note:
प्रत्येक
traverse
करने और
consideration
आना ना.

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

→ binary image.

Q2.) seed point = 6 (multiple থাকলে, pick any).
 $T = 3$ (diff ≤ 3)

5	6	6	7	6	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

max → 7 (min) → অন্য
 গাণ্ড না.

seed point 4 হলে,
 $2 \rightarrow 7 - 2 = 5 \geq 3$
 we will consider
 with region's max
 (Not with it's
 seed point).

segmented image →

1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	0	0	1	1
1	1	1	1			1	1
						1	1
						1	1
						1	1
						1	1

Region Growing :

Q1

	0	1	2	3
0	0	1	2	0
1	2	5	6	1
2	1	4	7	3
3	0	2	5	1

(if not given $\rightarrow (0,0)$)

initial point $(2,2)$

Threshold val = 2 ($\text{diff} \leq 2$)

4 connectivity $\rightarrow$ (only vertical, horizontal)

Highest $\rightarrow 7$

lowest $\rightarrow 5$

0	0	0	0
0	1	1	0
0	0	1	0
0	0	1	0

Q2

5	6	6	7	6	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	6	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

7 max in 7th row, 4 min in 4th row

$h \rightarrow 8, 7$ (max)

$l \rightarrow 8, 5, 4$ (min)

8 connectivity

$T = 3$ ($\text{diff} \leq 3$)

seed point = 6

प्रधान (याद) start

$\rightarrow$ circle it.

$\rightarrow$ all 1

$\rightarrow 1$

seed $\rightarrow 6$
 $T = 3$
 8 connectivity.

max $\rightarrow 9$
 min $\rightarrow 6$
 OK, max
 we have taken this one as 5

✓ 5	6	x 9	✓ 8	6	7	6	6	✓
✓ 6	✓ 7	✓ 5	✓ 7	✓ 5	5	x 4	7	✓
✓ 6	✓ 6	x 4	x 4	x 3	x 2	x 5	6	✓
✓ 5	x 4	✓ 5	x 4	2	x 3	x 4	✓ 6	
0	3	2	x 3	3	2	x 4	7	
0	0	✓ 6	0	2	2	✓ 5	6	
1	✓ 8	9	1	7	x 9	x 4	x 4	
1	0	1	0	2	3	5	4	

Note
 First traverse by left to right

max $\rightarrow 9$
 min $\rightarrow 6$

Then draw the segmented image.

Region Splitting :

→ whole image → 1 region

Q) Threshold val ≤ 4 ($T \leq 4$) diff > 4
square ना थाकल → make it square

Ex : $12 \times 7 \xrightarrow{\text{sqre}} 4 \times 4$

OR
 1×1 ✓

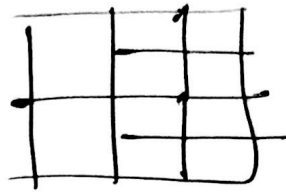
P
↑
 8×8 block → max → 7 > 4 जात split. (split in 4 subimage).
min → 0

	5	6	6	7	7	7	6	6	
Ⓐ	6	7	6	7	5	5	4	7	Ⓑ
	6	6	4	4	3	2	5	6	
	5	4	5	4	2	3	4	6	
P	0	3	2	3	3	2	4	7	
	0	0	0	0	2	2	5	6	
Ⓒ	1	1	0	1	0	3	4	4	Ⓓ
	1	0	1	0	2	3	5	4	

For subimage a → max = 7
min = 4 } No split.

b → max = 7
min = 2 } split into 4.

✓



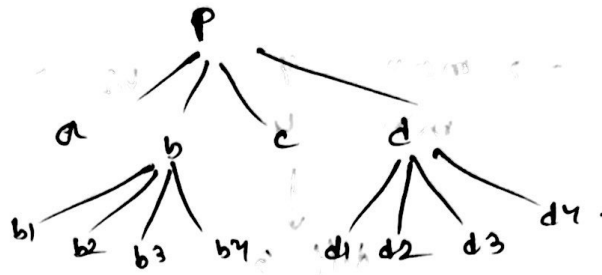
b1	7	7		6	6		b2
	5	5		4	7		
b3	3	2		5	6		b4
	2	3		4	6		

c → max 3
min 0 } No split.

d → max 7
min 0 } split in 4

d1	3	2		4	7		d2
	2	2		5	6		
d3	0	3		4	4		d4
	2	3		5	4		

- b1 → NS (7, 5)
- b2 → NS (7, 4)
- b3 → NS (3, 2)
- b4 → NS (6, 4)
- d1 → NS (3, 2)
- d2 → NS (7, 4)
- d3 → (3, 0) NS
- d4 → (5, 4) NS



Q3 $\rightarrow$ splitting + merging/growing.

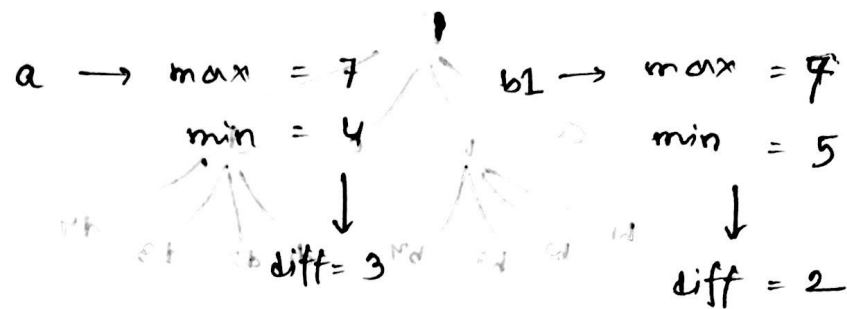
$$T = 3$$

motion
these
sub regions.

P	5	6	6	6	7	7	6	6	b
	6	7	6	7	5	5	4	7	
	6	6	4	4	3	2	5	6	
	5	4	5	4	2	3	4	6	
c	0	3	2	3	3	2	4	7	d
	0	0	0	0	2	2	5	6	
	1	1	0	1	0	3	4	4	
	1	0	1	0	2	3	5	4	

$p \rightarrow \begin{matrix} \max = 7 \\ \min = 0 \end{matrix} > \text{diff} = 7 > 3 \rightarrow \text{split in 4.}$

Note $\rightarrow$ split $\rightarrow$ 1 image draw } last 2
 merge $\rightarrow$ 1 " " }



a & b1 will be merged.

(a, b1) $\rightarrow$ max = 7
 min = 4

b2 $\rightarrow$ max = 7
 min = 4

a, b1 & b2 will be merged.

d3 $\rightarrow$ min = 2 X (min = 4)

d4 $\rightarrow$ min = 4 ✓

c $\rightarrow$ min = 0 X

d1 $\rightarrow$ min = 2 X

d2 $\rightarrow$ min = 4 ✓

d3 $\rightarrow$ min = 0 X

d4 $\rightarrow$ min = 4 ✓

Region Growing / Merging & Splitting :

5 ^x	8 [✓]	10 ^x	7 [✓]	8 [✓]	6 [✓]	9 [✓]	0 ^x
6 [✓]	7 [✓]	4 ^x	5 ^x	4 ^x	2 ^x	5 ^x	0 ^x
4 [✓]	6 [✓]	8 [✓]	3 ^x	6 [✓]	7 [✓]	6 [✓]	0 ^x
5 [✓]	2 [✓]	8 [✓]	6 [✓]	5 ^x	4 ^x	9 [✓]	0 ^x
2 [✓]	1 [✓]	3 ^x	0 ^x	2 ^x	5 ^x	4 ^x	0 ^x
0 [✓]	1 [✓]	2 [✓]	8 [✓]	6 [✓]	2 [✓]	4 [✓]	0 [✓]
2 [✓]	4 [✓]	3 [✓]	5 [✓]	9 [✓]	3 [✓]	3 [✓]	0 [✓]
0 [✓]	0 [✓]	0 [✓]	0 [✓]	0 [✓]	0 [✓]	0 [✓]	0 [✓]

seed point = 8

T = 3

max = 8

min = 0

5 [✓]	8 [✓]	10 ^x	7 [✓]	8 [✓]	6 [✓]	9 ^x	0 [✓]
6 [✓]	7 [✓]	4 ^x	5 ^x	4 ^x	2 ^x	5 ^x	0 [✓]
4 [✓]	6 [✓]	8 [✓]	3 ^x	6 [✓]	7 [✓]	6 [✓]	0 [✓]
5 [✓]	2 [✓]	8 [✓]	6 [✓]	5 ^x	4 ^x	9 ^x	0 [✓]
2 [✓]	1 [✓]	3 ^x	0 ^x	2 ^x	5 ^x	4 ^x	0 [✓]
0 [✓]	1 [✓]	2 [✓]	8 [✓]	6 [✓]	2 [✓]	4 [✓]	0 [✓]
2 [✓]	4 [✓]	3 [✓]	5 [✓]	9 [✓]	3 [✓]	3 [✓]	0 [✓]
0 [✓]	0 [✓]	0 [✓]	0 [✓]	0 [✓]	0 [✓]	0 [✓]	0 [✓]

max 8

min 5

4 region here

1. & 0 first segmentation.

1	1	0	1	1	1	0	0
1	1	0	1	0	0	1	0
0	1	1	0	1	1	1	0
1	0	1	1	1	0	0	0
0	0	0	0	0	1	0	0
0	0	0	1	1	0	0	0
0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0

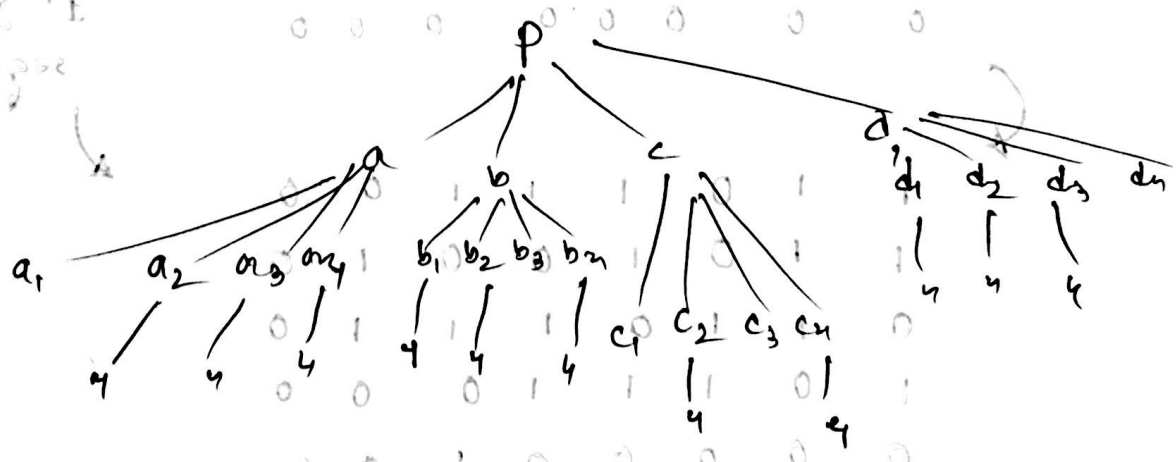
8 = 8
T = 3

a₁ 5 8 6 7				a₂ 10 7 4 5				b₁ 8 6 4 2				b₂ 9 0 5 0			
a₃ 4 6 5 2				a₄ 8 3 8 6				b₃ 6 7 5 4				b₄ 6 0 9 0			
c₁ 2 1 0 1				c₂ 3 0 2 8				d₁ 2 5 6 2				d₂ 4 0 4 0			
c₃ 0 0 0 0				c₄ 0 0 0 0				d₃ 9 3 0 0				d₄ 3 0 0 0			

P → max = 10
min = 0 } split

a → max = 10
min = 2 } split

a₁ → 8, 5 ✓ No split.
a₂ → 10, 4 → split.



merge it

Region Growing :

2	3	4 ^x	7	9	8
1	0	4 ^x	7	8	6
0	0	5	9	5	11 ^x
3	3	6	8	3	4
5	2	9	4	0	5
11	3	6	1	5	8
16	2	10 ^x	8	7	9
/	/	/	/	/	/

seed $\rightarrow$ 6

Threshold = 3

4 adjacency.

max 8

min 5

Region Splitting :

8x8 image

cropped $\rightarrow$

6	8	5	9	6	8	9	0
8	6	4	5	3	7	4	0
9	4	5	2	1	2	5	0
3	4	2	2	0	3	4	0
0	3	4	5	4	4	5	0
0	5	2	0	3	8	7	0
2	9	3	2	3	3	6	0
5	1	0	3	8	2	4	0
0	0	2	1	9	0	3	0

zero padding

$\rightarrow 9 \times 7$

resize to 8x8

crop / padding

be clever while cropping.

(seed point not part crop).

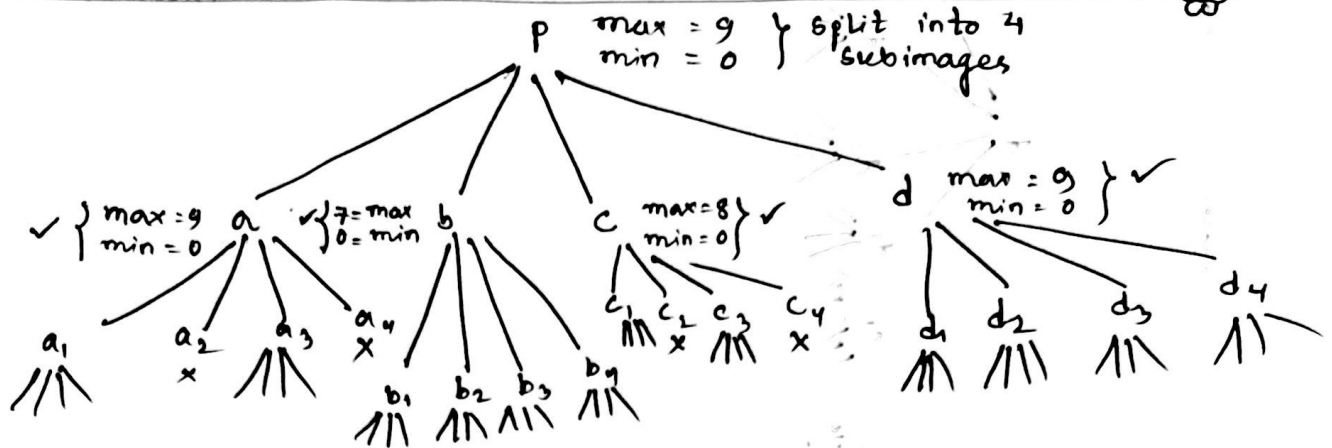
now split & merge.

seed 6

T $\rightarrow$ 3

P $\rightarrow$ max = 9
min = 0 $\rightarrow$ diff = 9

split into 4 subsection $\rightarrow$ ✓
 otherwise $\rightarrow$ x

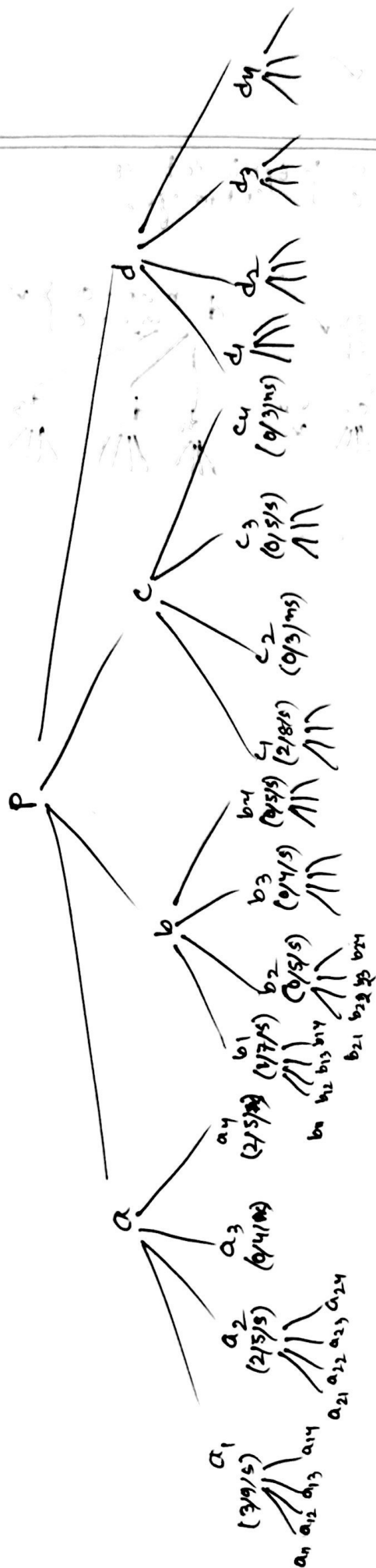


$a_3 \rightarrow$ merge

$a_3, a_4 \rightarrow$ merge

$c_3 \rightarrow$ merge

$c_3, c_4 \rightarrow$ merge



To merge
min $\rightarrow 3$
max $\rightarrow 6$

3 to 6 $\rightarrow T=3$

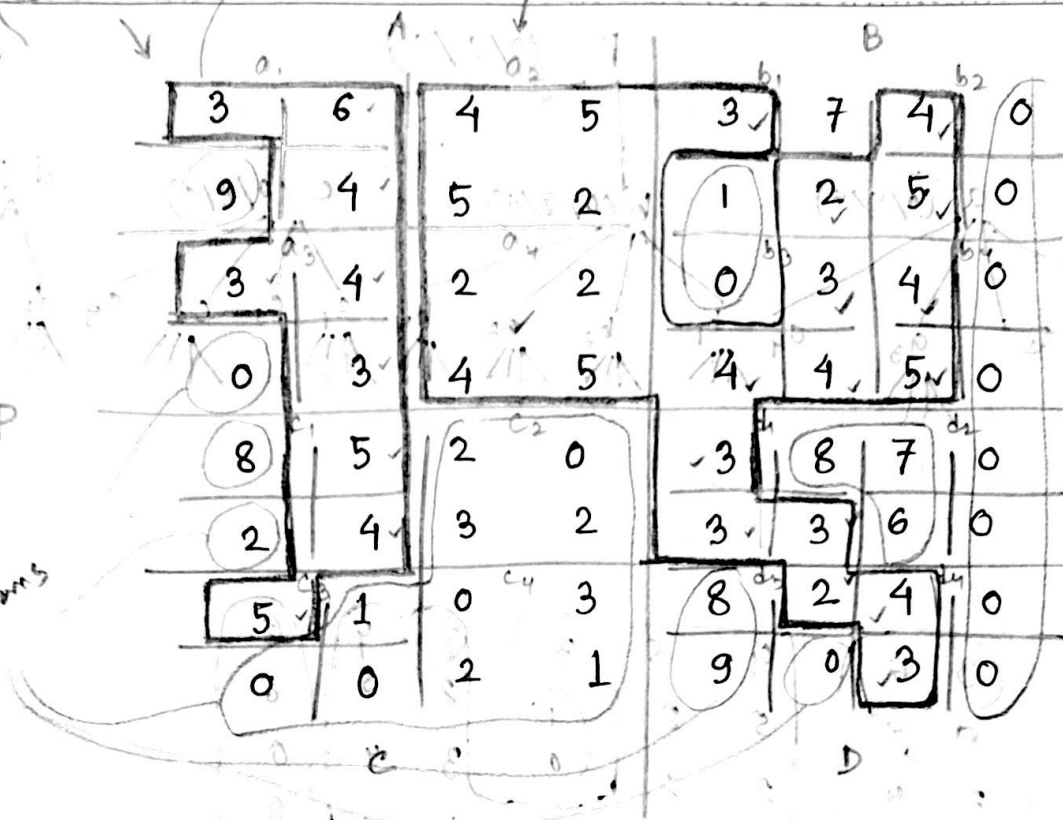
min=2
max=4

Split & merge

↓
Resize to
8x8

$T \rightarrow 3$

P
orphans

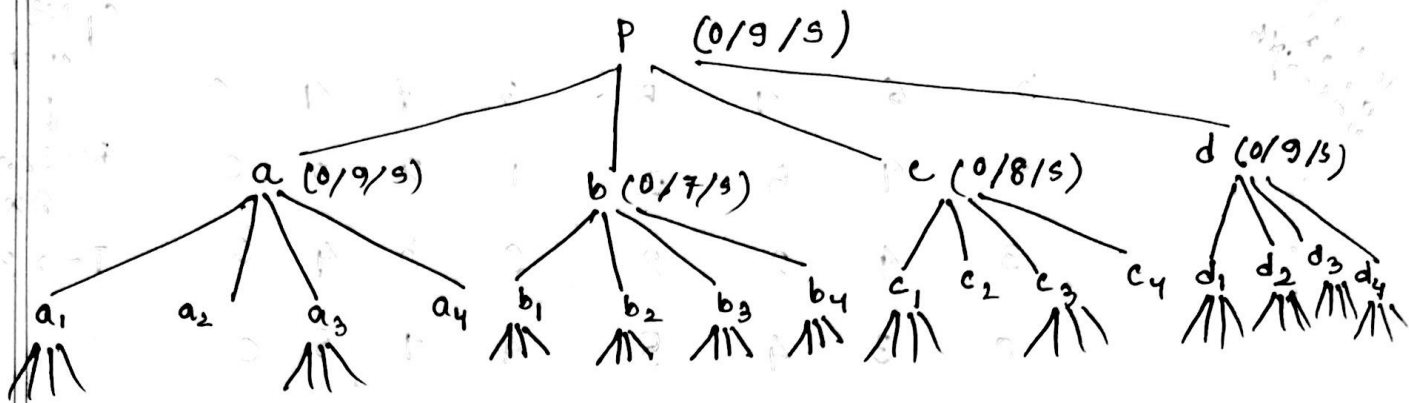


To resize $\rightarrow$ left/right & zero padding
or mirror padding.

let. image mask $\rightarrow P$

split in 4 blocks $\rightarrow S$

no split $\rightarrow ns$.



CT 2 syllabus :

Region Growing , splitting , merging
(related all theories)

Note .

6	10	9	70	75	85	48
68	5	8	4	79	23	45
0	60	31	39	25	21	40
70	0	64	200	194	24	57
254	250	202	207	210	55	1
251	99	205	210	54	0	0
101	102	98	59	1	0	2
254	100	61	5	2	3	4

perform
split & merge

$T \rightarrow 20$

Note :

Do not
reduce the
size of the
image

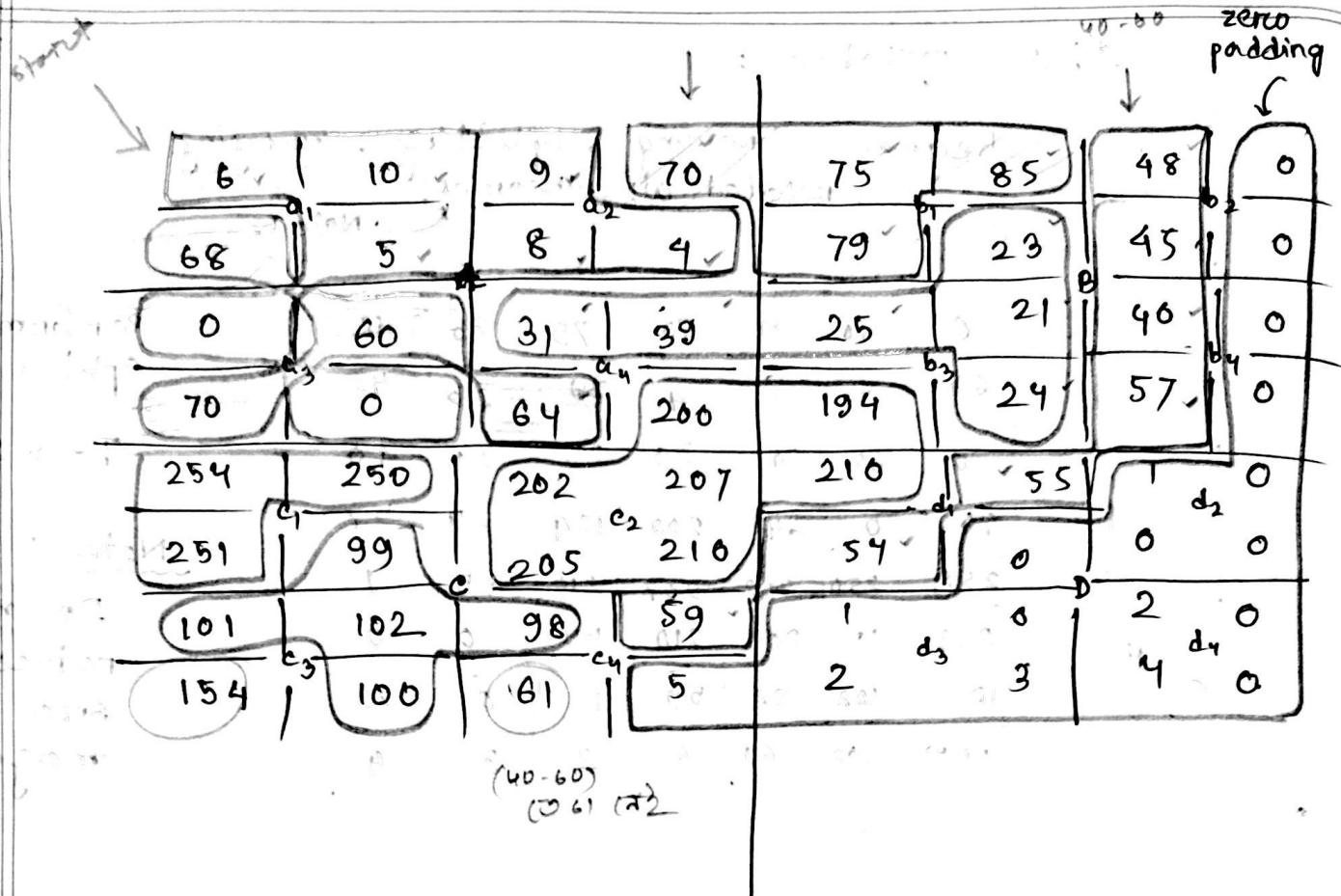
8x7 image size

For splitting , we need square block. ($2^m \times 2^n$)
make it to 8x8 / 16x16

use zero padding at right.

⇒ For splitting image size must be $m \times n$ (where m & n must be 2^n).

Resize the image block as 8x8 while adding
right col with zero padding.



original image → P

Max intensity → M_1

Min " → M_2

Split into 4 blocks → S

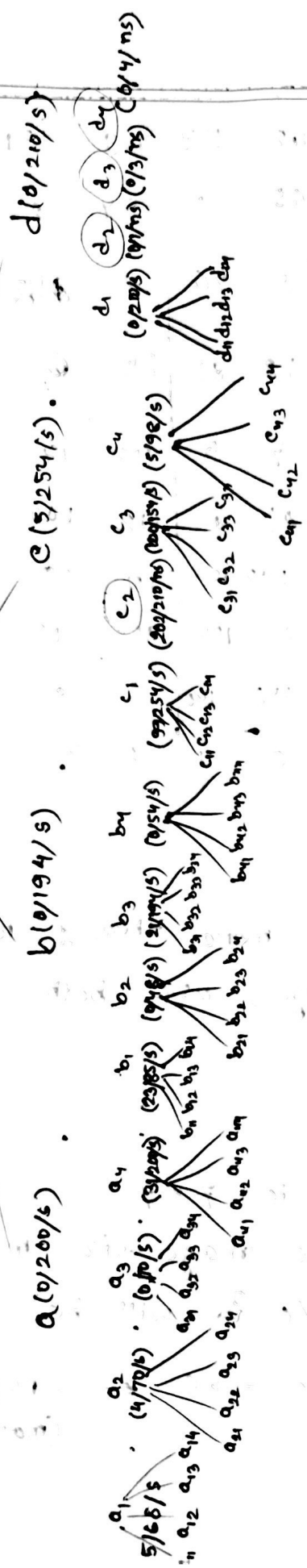
No split → ns

$M_1 > \text{diff} > 20 \rightarrow S$

$M_2 > \text{diff} \leq 20 \rightarrow ns.$

g/ms

M_1 M_2 ↑
P (0/254/s)



0	15	19	33	47	42	127	129	131	141
70	61	40	87	31	65	9	84	2	71
72	85	89	42	86	5	70	3	142	1
75	67	45	85	37	72	81	75	113	
101	55	0	9	13	7	5	110	114	115
105	53	1	11	19	13	4	112	100	120
110	215	42	0	145	133	135	14	19	17
200	202 3	203	254	137	5	134	11	13	12
205	218 5	207	188	139	142	140	5	9	7
255	201	195	99	0	0	1	2	3	4

10x10
image
7x7
image

Split & merge $T \rightarrow 20$

Do not reduce the size if more than 1 r/c

addition to the org. image, use both left & right / top & bottom

→ Given image (let P) size is 7×7 .

For splitting operation, the image size must have to be square where r/c & will be 2^n .

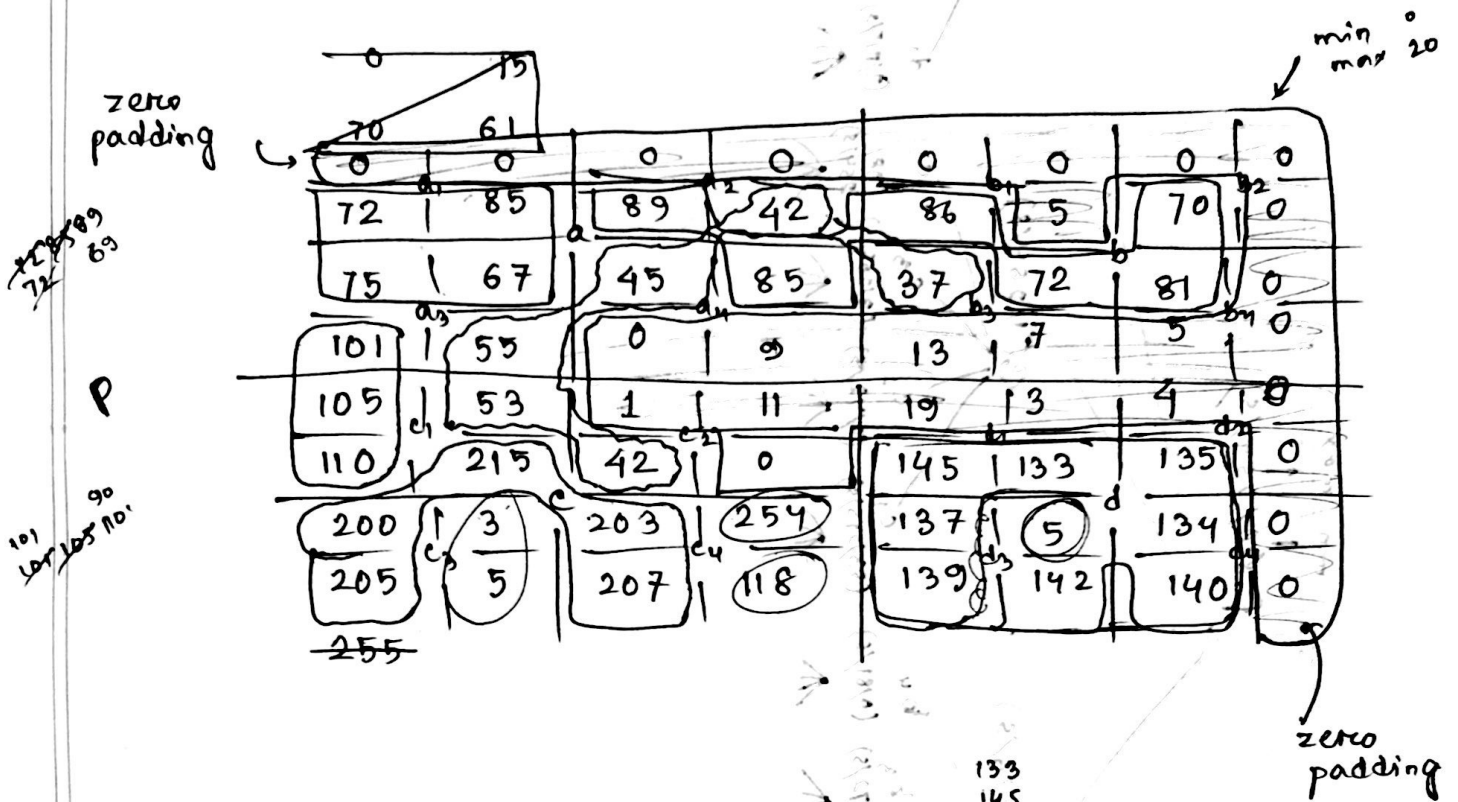
let, split into 4 subimage $\rightarrow S$ ← region diff

No split $\rightarrow ns$

(max-min) > 20

Region (min/max) diff $> T \rightarrow S$
 else, ns

① create square image of 8×8 with zero padding



min max

$P(0/254/5)$

